

Homework 4

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Problem 1 Prove that the map

$$S^{4n-1} \rightarrow S^{2n}$$

defined by the Whitehead square of the generator has Hopf invariant 2.

Solution: The mapping cone of the Whitehead bracket $S^{2n-1} \rightarrow S^n$ is $C_{[\iota, \iota]} = S^n \cup_{[\iota, \iota]} e^{2n}$ the space obtained by folding $S^n \vee S^n \subset S^n \times S^n$ to a single S^n . We have

$$H^n(S^n \times S^n) = \mathbb{Z}a \oplus \mathbb{Z}b,$$

and

$$H^{2n}(S^n \times S^n) = \mathbb{Z}ab.$$

Observe that the class of the folded sphere $x \in H^n(C_{[\iota, \iota]}, \mathbb{Z}) \cong \mathbb{Z}$ pulls back to $a + b$ in $S^n \times S^n$, and consequently x^2 goes to $(a + b)^2$. Now,

$$(a + b)^2 = a^2 + b^2 + ab + ba = ab + ba = (1 + (-1)^n)ab.$$

In particular, the Hopf invariant is

$$H([\iota, \iota]) = \begin{cases} 2 & \text{if } n \text{ is even,} \\ 0 & \text{if } n \text{ is odd.} \end{cases}$$

Problem 2 (Hatcher AT, 4.2.34) Let $p : S^3 \rightarrow S^2$ be the Hopf bundle and $q : T^3 \rightarrow S^3$ be the quotient map collapsing the complement of a ball in the 3-dimensional torus $T^3 = S^1 \times S^1 \times S^1$ to a point. Show that $pq : T^3 \rightarrow S^2$ induces the zero map on homotopy and reduced homology, but is *not* homotopic to a constant map.

Solution: Let $p : S^3 \rightarrow S^2$ be the Hopf fibration and let

$$q : T^3 \rightarrow S^3$$

be the quotient collapsing the complement of an embedded 3-ball in $T^3 = S^1 \times S^1 \times S^1$ to a point. Set $f := p \circ q$.

The cellular approximation theorem allows us to homotope q to a cellular map, which we also denote by q . This q must collapse the 2-skeleton of T^3 a point and so $q_* = 0$ for $i < 3$. Also q_* is an isomorphism in degree 3 since it sends the fundamental class of the chosen ball to the fundamental class of S^3 . On the other hand, p_* is 0 in degree 3 since $H_3(S^2) = 0$. Therefore, the composition $f = p \circ q$ induces the zero map on reduced homology.

The universal cover of T^3 is $\mathbb{R}^3$, which is contractible, so $\pi_i(T^3) = 0$ for $i > 1$. Also $\pi_1(T^3) = \mathbb{Z}^3$ and $\pi_1(S^2) = 0$ so f_* is the zero map on all homotopy groups.

Suppose for contradiction that f is homotopic to a constant map. Then there exists a homotopy

$$H : T^3 \times I \rightarrow S^2$$

with $H(-, 0) = f$ and $H(-, 1) = c$ constant.

Since $p : S^3 \rightarrow S^2$ is a fibration and q is a lift of f , the homotopy lifting property produces a lift

$$\tilde{H} : T^3 \times I \rightarrow S^3$$

such that

$$\tilde{H}(-, 0) = q \quad \text{and} \quad p \circ \tilde{H} = H.$$

At time $t = 1$, since $H(-, 1) = c$ is constant, the map $\tilde{H}(-, 1)$ must land in a single fiber of p , which is a circle $S^1 \subset S^3$. Thus $\tilde{H}(-, 1)$ factors as

$$T^3 \longrightarrow S^1 \hookrightarrow S^3.$$

Therefore q is homotopic (as a map into S^3) to a map factoring through S^1 . But this is impossible: the map q has degree 1, so

$$q_* : H_3(T^3) \rightarrow H_3(S^3)$$

is an isomorphism, whereas any map $T^3 \rightarrow S^1 \subset S^3$ induces the zero map on H_3 since $H_3(S^1) = 0$.

Problem 3 For a relatively finite CW pair (X, A) (meaning that X is built by attaching finitely many cells to A) and any space Y , show that the restriction map

$$\text{Map}(X, Y) \rightarrow \text{Map}(A, Y)$$

is a fibration.

Solution: Let

$$r : \text{Map}(X, Y) \rightarrow \text{Map}(A, Y)$$

be the restriction map. We show that r is a fibration.

It suffices to verify the homotopy lifting property with respect to disks. So suppose we are given a commutative diagram

$$\begin{array}{ccc} D^n & \xrightarrow{g} & \text{Map}(X, Y) \\ \downarrow & & \downarrow r \\ D^n \times I & \xrightarrow{H} & \text{Map}(A, Y) \end{array}$$

where $H(u, 0) = r(g(u))$ for all $u \in D^n$. We must construct a lift

$$\tilde{H} : D^n \times I \rightarrow \text{Map}(X, Y)$$

such that $\tilde{H}(u, 0) = g(u)$ and $r \circ \tilde{H} = H$.

By adjunction, g corresponds to a map

$$\hat{g} : D^n \times X \rightarrow Y,$$

and H corresponds to a map

$$\hat{H} : D^n \times I \times A \rightarrow Y.$$

The condition $H(u, 0) = r(g(u))$ says exactly that

$$\hat{H}(u, 0, a) = \hat{g}(u, a) \quad \text{for all } u \in D^n, a \in A.$$

Hence $\hat{g}$ and $\hat{H}$ together define a map

$$F : (D^n \times X \times \{0\}) \cup (D^n \times A \times I) \rightarrow Y.$$

So it remains to extend F to all of $D^n \times X \times I$.

Now (X, A) is a relatively finite CW pair, so X is obtained from A by attaching finitely many cells. Therefore $(D^n \times X, D^n \times A)$ is again a relative CW pair, hence the inclusion

$$D^n \times A \hookrightarrow D^n \times X$$

is a cofibration. It follows that the inclusion

$$(D^n \times X \times \{0\}) \cup (D^n \times A \times I) \hookrightarrow D^n \times X \times I$$

is also a cofibration, so the homotopy extension property applies. Thus F extends to a map

$$\tilde{F} : D^n \times X \times I \rightarrow Y.$$

Adjointing back, $\tilde{F}$ corresponds to a map

$$\tilde{H} : D^n \times I \rightarrow \text{Map}(X, Y)$$

with

$$\tilde{H}(u, 0) = g(u) \quad \text{and} \quad r(\tilde{H}(u, t)) = H(u, t).$$

Thus r has the homotopy lifting property for disks, hence is a fibration.

Problem 4 Given fibrations $E, E' \rightarrow B$, show that any fiber-preserving map

$$f : E \rightarrow E'$$

which is a homotopy equivalence (on total spaces) is a *fiber homotopy equivalence*, meaning there exists a fiber-preserving map

$$g : E' \rightarrow E$$

such that $g \circ f$ and $f \circ g$ are both homotopic to the identities via fiber-preserving maps.

Solution: We first record a standard lemma.

Lemma. Let $p : E \rightarrow B$ be a fibration, and let $u : X \rightarrow B$ be a map. If

$$f_0, f_1 : X \rightarrow E$$

satisfy

$$p \circ f_0 = p \circ f_1 = u,$$

and if $f_0 \simeq f_1$ as ordinary maps $X \rightarrow E$, then in fact f_0 and f_1 are homotopic through fiber-preserving maps, i.e. there exists

$$\tilde{H} : X \times I \rightarrow E$$

with

$$\tilde{H}(-, 0) = f_0, \quad \tilde{H}(-, 1) = f_1, \quad p\tilde{H}(x, t) = u(x).$$

Proof. Let

$$H : X \times I \rightarrow E$$

be a homotopy from f_0 to f_1 . Its projection

$$pH : X \times I \rightarrow B$$

is a homotopy from u to u . Define

$$G : X \times I \times I \rightarrow B, \quad G(x, s, t) = pH(x, (1-s)t).$$

Then

$$G(x, 0, t) = pH(x, t), \quad G(x, 1, t) = u(x).$$

So G is a homotopy, relative $X \times \{0, 1\}$, from pH to the constant homotopy at u .

Now define

$$\Phi : X \times I \times \{0\} \cup X \times \{0\} \times I \cup X \times \{1\} \times I \rightarrow E$$

by

$$\Phi(x, s, 0) = f_0(x), \quad \Phi(x, 0, t) = H(x, t), \quad \Phi(x, 1, t) = f_1(x).$$

These formulas agree on overlaps, and one checks that $p\Phi = G$ on this subspace. Since

$$(X \times I \times \{0\}) \cup (X \times \{0, 1\} \times I) \hookrightarrow X \times I \times I$$

is a cofibration and p is a fibration, the homotopy lifting property yields an extension

$$\tilde{\Phi} : X \times I \times I \rightarrow E$$

with $p\tilde{\Phi} = G$. Set $\tilde{H}(x, t) := \tilde{\Phi}(x, 1, t)$. Then $\tilde{H}(-, 0) = f_0$, $\tilde{H}(-, 1) = f_1$ and

$$p\tilde{H}(x, t) = G(x, 1, t) = u(x)$$

for all (x, t) . Thus $\tilde{H}$ is a fiber-preserving homotopy. $\square$

Now let

$$p : E \rightarrow B, \quad p' : E' \rightarrow B$$

be fibrations, and let

$$f : E \rightarrow E'$$

be fiber-preserving, so $p'f = p$. Assume f is a homotopy equivalence of total spaces. Choose an ordinary homotopy inverse

$$h : E' \rightarrow E$$

with

$$hf \simeq \text{id}_E, \quad fh \simeq \text{id}_{E'}.$$

We first deform h to a fiber-preserving map. Since $p'f \simeq p' \circ \text{id}_{E'}$ and $fh \simeq \text{id}_{E'}$, we have

$$ph = p'fh \simeq p'.$$

Apply the homotopy lifting property to the homotopy $ph \simeq p'$, starting with the map h . This gives a homotopy from h to a map

$$g : E' \rightarrow E$$

such that

$$pg = p'.$$

So g is fiber-preserving.

Now $g \simeq h$, hence

$$gf \simeq hf \simeq \text{id}_E$$

and

$$fg \simeq fh \simeq \text{id}_{E'}.$$

Moreover both pairs of maps are fiber-preserving:

$$p(gf) = p'f = p = p(\text{id}_E),$$

$$p'(fg) = p'g = p' = p'(\text{id}_{E'}).$$

Therefore, by the lemma, these ordinary homotopies can be replaced by fiber-preserving homotopies. Thus

$$gf \simeq_B \text{id}_E, \quad fg \simeq_B \text{id}_{E'}.$$

Hence g is a fiber-preserving homotopy inverse to f , so f is a fiber homotopy equivalence.

Problem 5 Show that a fibration over a *contractible* base is fiber homotopy equivalent to a product space. Show the same for a fibration with contractible fiber with CW base. Assuming that a fiber bundle over a paracompact base is a fibration (for example when the fibers are smooth manifolds), conclude that every fiber bundle over a contractible paracompact base is a product bundle.

Solution: First suppose $p : E \rightarrow B$ is a fibration and B is contractible. Choose a basepoint $b_0 \in B$, let

$$F = p^{-1}(b_0),$$

and let

$$H : B \times I \rightarrow B$$

be a contraction of B to b_0 , so $H(b, 0) = b$ and $H(b, 1) = b_0$.

Now H is a homotopy from id_B to the constant map $c : B \rightarrow B$, $c(b) = b_0$. Pulling back the fibration p along these two maps gives:

$$(\text{id}_B)^*E \cong E, \quad c^*E \cong B \times F.$$

Since pullbacks of a fibration along homotopic maps are fiber homotopy equivalent, it follows that

$$E \simeq_B B \times F.$$

Thus a fibration over a contractible base is fiber homotopy equivalent to a product.

Now suppose instead that $p : E \rightarrow B$ has contractible fiber F . We claim that p is a fiber homotopy equivalence with inverse given by a section $s : B \rightarrow E$. Indeed, over a CW decomposition of B , the obstruction to extending a section over an n -cell lies in $\pi_{n-1}(F)$, and these groups vanish since F is contractible. Hence a section s exists. Then

$$p \circ s = \text{id}_B.$$

Also $s \circ p : E \rightarrow E$ is fiber-preserving, and on each fiber E_b it is the constant map to $s(b)$. Since each fiber is contractible, $s \circ p$ is fiberwise homotopic to the identity. Therefore

$$s \circ p \simeq_B \text{id}_E,$$

so p is a fiber homotopy equivalence. Hence E is fiber homotopy equivalent to B .

Finally, let $p : E \rightarrow B$ be a fiber bundle over a contractible paracompact base B , and choose $b_0 \in B$. Let $c : B \rightarrow B$ be the constant map $c(b) = b_0$. Since B is contractible, id_B is homotopic to c . By the bundle homotopy theorem for bundles over paracompact bases, pullbacks along homotopic maps are isomorphic. Hence

$$E \cong \text{id}_B^*E \cong c^*E.$$

If $F = p^{-1}(b_0)$, then $c^*E \cong B \times F$. Therefore

$$E \cong B \times F$$

as bundles over B . Thus every fiber bundle over a contractible paracompact base is a product bundle.

Problem 6 (Hatcher AT, 4.2.17) Show that the map

$$[X, Y] \rightarrow \text{Hom}(\pi_n X, \pi_n Y)$$

sending $[f] \mapsto f_*$ is a bijection, if X is an $(n-1)$ -connected CW complex and Y is a path-connected space with $\pi_i Y = 0$ for $i > n$. Deduce that CW models for $K(G, n)$ are unique up to homotopy equivalence.

Solution: Since X is $(n-1)$ -connected, we may choose a CW structure on X with

$$X^{n-1} = *.$$

Then X^n is a wedge of n -spheres,

$$X^n \cong \bigvee_{\alpha} S_{\alpha}^n,$$

and the quotient X^n/X^{n-1} is the same wedge.

A based map

$$f_n : X^n \rightarrow Y$$

is therefore determined up to homotopy by its restrictions to the n -cells, hence by a cellular n -cochain with values in $\pi_n(Y)$. More concretely, such a map determines a homomorphism

$$C_n(X) \rightarrow \pi_n(Y),$$

and two such maps are homotopic rel X^{n-1} exactly when they induce the same homomorphism

$$H_n(X) \rightarrow \pi_n(Y).$$

Thus homotopy classes of maps $X^n \rightarrow Y$ correspond naturally to

$$\text{Hom}(H_n(X), \pi_n(Y)).$$

Now we claim that every map $X^n \rightarrow Y$ extends over all higher skeleta of X , and that any two such extensions are homotopic. Indeed, the obstruction to extending from X^k to X^{k+1} lies in

$$H^{k+1}(X; \pi_k(Y)).$$

But for $k \geq n$, we have $\pi_k(Y) = 0$ unless $k = n$, and for $k = n$ the map is already defined on X^n . Hence all higher obstructions vanish. Similarly, the obstruction groups controlling uniqueness of extension involve $\pi_k(Y)$ for $k > n$, so they also vanish. Therefore restriction induces a bijection

$$[X, Y] \xrightarrow{\cong} [X^n, Y].$$

Combining this with the previous paragraph gives

$$[X, Y] \cong \text{Hom}(H_n(X), \pi_n(Y)).$$

Since X is $(n-1)$ -connected, the Hurewicz theorem gives

$$\pi_n(X) \cong H_n(X),$$

so we obtain a natural bijection

$$[X, Y] \cong \text{Hom}(\pi_n(X), \pi_n(Y)).$$

It remains to note that this bijection is exactly the map $[f] \mapsto f_*$. Indeed, if $f : X \rightarrow Y$, then its restriction to X^n determines the induced homomorphism

$$H_n(X) \rightarrow \pi_n(Y),$$

and under the Hurewicz isomorphism $H_n(X) \cong \pi_n(X)$, this is precisely the induced map

$$f_* : \pi_n(X) \rightarrow \pi_n(Y).$$

Thus the stated map is a bijection.

Now let X and Y be CW models for $K(G, n)$. Then X is $(n-1)$ -connected and Y is path-connected with $\pi_i(Y) = 0$ for $i > n$, so the first part applies:

$$[X, Y] \cong \text{Hom}(\pi_n(X), \pi_n(Y)).$$

Since

$$\pi_n(X) \cong G \cong \pi_n(Y),$$

the identity map on G is induced by some map

$$f : X \rightarrow Y.$$

Similarly there is a map

$$g : Y \rightarrow X$$

inducing the identity on π_n . Then

$$(g \circ f)_* = \text{id}_{\pi_n(X)}, \quad (f \circ g)_* = \text{id}_{\pi_n(Y)}.$$

Applying the bijection again, we conclude that

$$g \circ f \simeq \text{id}_X, \quad f \circ g \simeq \text{id}_Y.$$

Hence f and g are homotopy inverses. Therefore CW models for $K(G, n)$ are unique up to homotopy equivalence.

Problem 7 (var. of Hatcher AT, 4.3.6 & 4.3.7) For an Abelian group G , Question 6 gives a map

$$\mu : K(G, n) \times K(G, n) \rightarrow K(G, n)$$

corresponding to the addition on G . Show that μ defines an H -space structure on $K(G, n)$ that is commutative and associative up to homotopy.

For any CW space X , use μ to define an addition on $[X, K(G, n)]$ by

$$[f] + [g] := \mu(f \times g)$$

and show that under the bijection

$$[X, K(G, n)] \cong H^n(X; G)$$

this corresponds to the usual addition in cohomology.

Solution: Let $Y = K(G, n)$. Let $0 \in G$ denote the identity element, and let $*$ $\in Y$ be the basepoint. The constant map

$$e : * \rightarrow Y$$

corresponds to $0 \in \pi_n(Y) = G$. Consider the maps

$$\mu \circ (\text{id}, e), \quad \mu \circ (e, \text{id}) : Y \rightarrow Y.$$

On $\pi_n(Y) \cong G$, both induce the homomorphism

$$a \mapsto a + 0 = a.$$

By Question 6, these maps are homotopic to id_Y . Thus $*$ is a homotopy identity for μ , so (Y, μ) is an H -space. To show commutativity, let

$$\tau : Y \times Y \rightarrow Y \times Y, \quad \tau(x, y) = (y, x).$$

Then the two maps

$$\mu, \quad \mu \circ \tau : Y \times Y \rightarrow Y$$

induce on $\pi_n(Y \times Y) \cong G \oplus G$ the homomorphisms

$$(a, b) \mapsto a + b, \quad (a, b) \mapsto b + a.$$

Since G is Abelian, these are equal. Hence by Question 6,

$$\mu \circ \tau \simeq \mu.$$

So μ is commutative up to homotopy. To show associativity, consider the two maps

$$\mu \circ (\mu \times \text{id}), \quad \mu \circ (\text{id} \times \mu) : Y \times Y \times Y \rightarrow Y.$$

Since

$$\pi_n(Y \times Y \times Y) \cong G \oplus G \oplus G,$$

the first induces

$$(a, b, c) \mapsto (a + b) + c,$$

while the second induces

$$(a, b, c) \mapsto a + (b + c).$$

These are equal because addition in G is associative. Therefore, again by Question 6,

$$\mu \circ (\mu \times \text{id}) \simeq \mu \circ (\text{id} \times \mu).$$

So μ is associative up to homotopy. Now let X be any CW complex. For maps $f, g : X \rightarrow Y$, define

$$[f] + [g] := [\mu \circ (f, g)],$$

where

$$(f, g) : X \rightarrow Y \times Y, \quad x \mapsto (f(x), g(x)).$$

This gives an addition on $[X, Y]$.

We now show that under the canonical bijection

$$[X, K(G, n)] \cong H^n(X; G),$$

this addition agrees with the usual addition in cohomology. Let $\alpha, \beta \in H^n(X; G)$ correspond to $f, g : X \rightarrow K(G, n)$. By naturality of the representing-space correspondence, the product map $(f, g) : X \rightarrow K(G, n) \times K(G, n)$ corresponds to the pair $(\alpha, \beta) \in H^n(X; G) \oplus H^n(X; G)$. The map $\mu : K(G, n) \times K(G, n) \rightarrow K(G, n)$ was chosen to represent the addition map $G \oplus G \rightarrow G$, so $\mu \circ (f, g)$ corresponds to the sum $\alpha + \beta \in H^n(X; G)$.